Visualization Course Final Project (Fall 2024)

Amazon Dashboard

Made by :

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Project Details:

Title of the Client(s):

- Data Analysts
- Sales Teams
- Product Managers

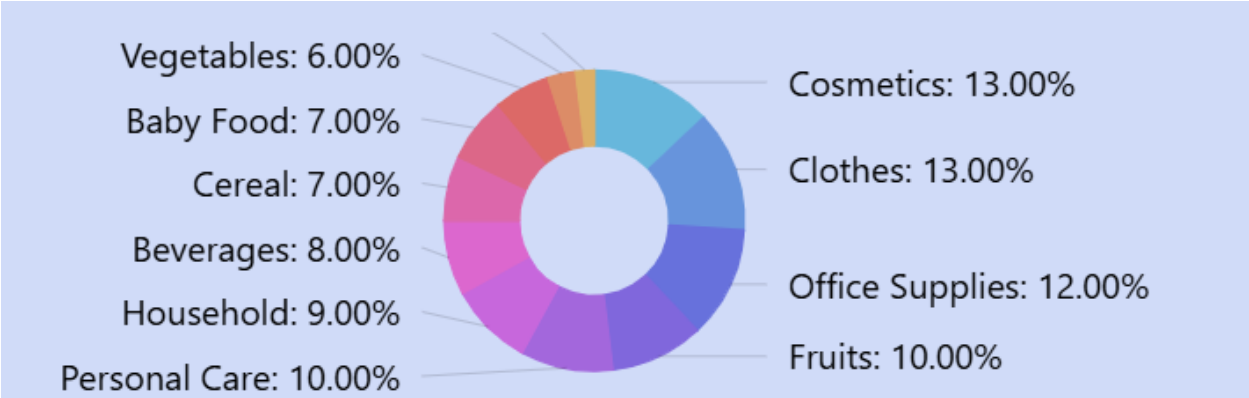
Available Information Provided by the Client:

- Product Categories (e.g., Vegetables, Baby Food, Clothes, etc.)
- Sales Data (Offline vs. Online percentages)
- Regional Sales Distribution (North America, Sub-Saharan Africa, etc.)
- Product Distribution (e.g., Cosmetics, Fruits, etc.)
- Sales Volume by Category

Required Questions for the Dashboard:

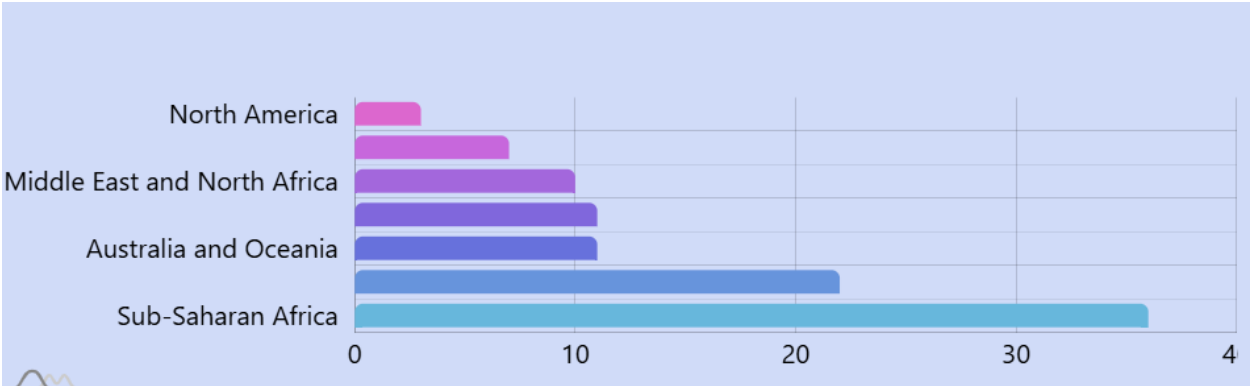
- What percentage of sales is offline vs. online?
- What are the sales distributions across different regions?
- How are product categories distributed in sales?
- Which product categories have the highest and lowest sales?

II. Chart Details



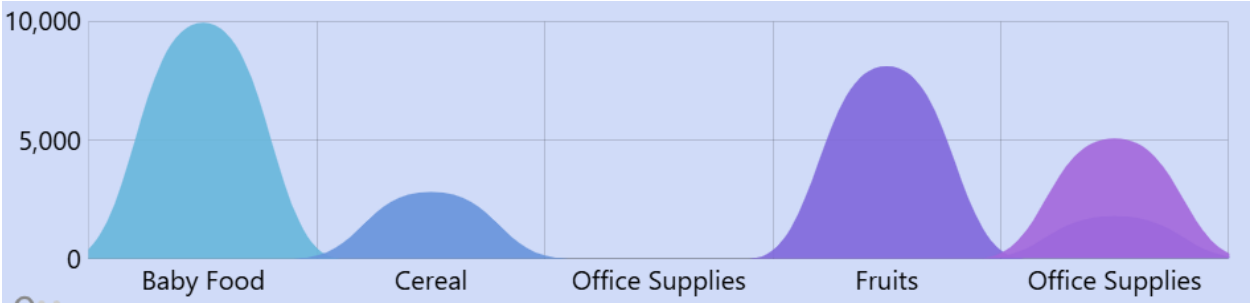
1. Donut Chart

- **Details:**
  - **Chart Type:** Donut Chart
  - **Title:** "Product Category Distribution"
  - **Colors:** Distinct colors for each product category (e.g., purple for Personal Care, blue for Cosmetics).
  - **Legend:** Yes, with category names and percentages.
- **Questions Answered:**
  - What is the percentage distribution of each product category?
- **Visibility:** Always visible on the dashboard.
- **Reason for Selection:**
  - Donut charts provide a quick overview of percentage distribution, making it easy to compare multiple categories.



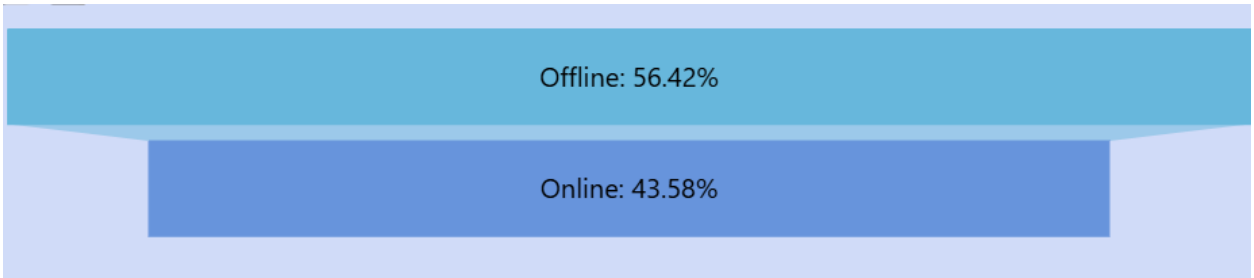
2. Bar Chart

- **Details:**
  - **Chart Type**: Horizontal Bar Chart
  - **Title**: "Regional Sales Distribution"
  - **Colors**: Gradient colors for better visual differentiation of regions.
  - **Axis Titles:**
    - X-Axis: Sales Percentage
    - Y-Axis: Regions
  - Legend: None required.
- **Questions Answered:**
  - How are sales distributed across different regions?
  - Which regions contribute the most and least to sales?
- **Visibility**: Always visible on the dashboard.
- **Reason for Selection:**
  - Horizontal bar charts are ideal for comparing values across regions.



3. **Area Chart**

- **Details:**
  - **Chart Type:** Stacked Area Chart
  - **Title**: "Sales Volume by Product Categories"
  - **Colors:** Muted gradients to separate categories visually.
  - **Axis Titles:**
    - X-Axis: Product Categories
    - Y-Axis: Sales Volume
  - **Legend:** Yes, with product names.
- **Questions Answered:**
  - Which product categories have the highest sales volume?
- **Visibility:** Always visible.
- **Reason for Selection:**
  - Stacked area charts provide a layered view of multiple variables, making it easier to see overall trends.



4. **Horizontal Bar (Stacked)**

- **Details:**
  - **Chart Type:** Stacked Horizontal Bar
  - **Title:** "Sales: Offline vs. Online"
  - **Colors:** Light blue for Offline, Dark Blue for Online.
  - **Legend:** Yes, indicating Offline and Online.
- **Questions Answered:**
  - What is the percentage of online vs. offline sales?
- **Visibility:** Always visible.
- **Reason for Selection:**
  - Stacked bars are excellent for showing proportions of two categories (e.g., offline vs. online).
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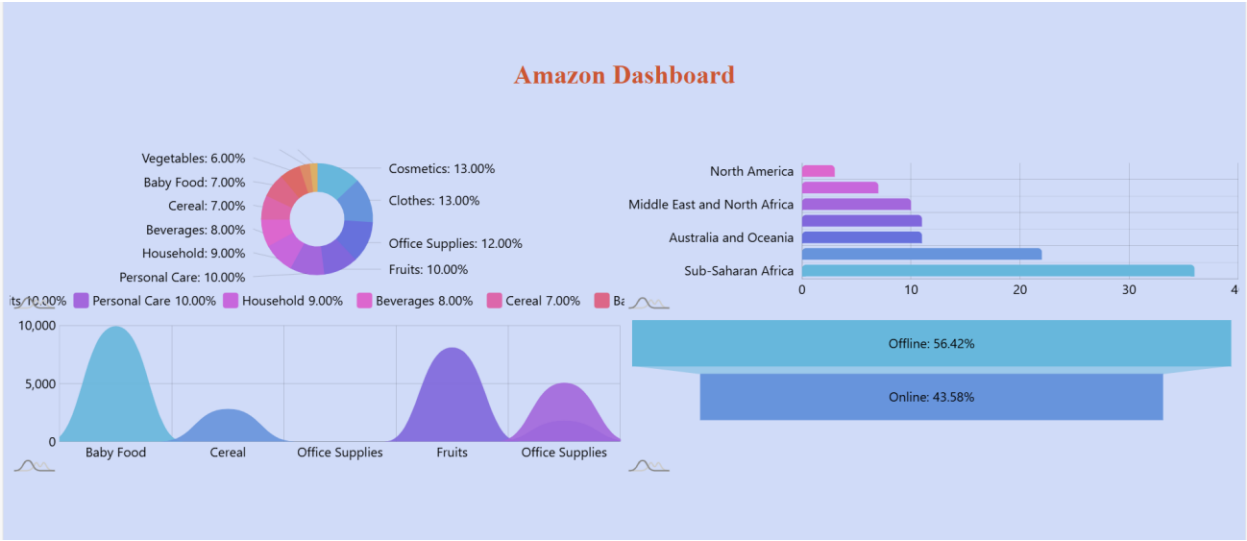
III. Dashboard Layout

a. Layout Explanation

1. **Title Position:**
    - Placed at the top center to immediately inform users about the dashboard purpose.
2. **Donut Chart Position:**
    - Positioned on the left for quick access to product category insights.
3. **Bar Chart Position:**
    - Positioned on the top-right for easy comparison of regional data.
4. **Area Chart Position:**
    - Positioned below the donut chart for seamless trend analysis.
5. **Stacked Bar Chart Position:**
    - Positioned at the bottom for summarizing online vs. offline sales.

b. Potential Criticism Points

1. **Cluttered Layout:**
    - Some users may find the layout crowded if too many charts are displayed simultaneously.
2. **Color Overlap:**
    - Similar colors in different charts might confuse users.
3. **Scalability:**
    - Adding new charts or data might disrupt the current layout.



Suggested Future Work

- Integrate real-time data updates from Amazon's API.
- Add filters and interactive features for dynamic data analysis.
- Improve scalability by introducing a dashboard customization option.
- Enhance performance with optimized database queries.
- Add more charts.

The data provided:

Amazon.csv file.